

ARKOPROBHO MITRA

AI Systems Engineer | Production GenAI Platforms | Technical Leadership

arko.mitra223@gmail.com | +91 79084 91618 | Kolkata, India

Portfolio: portfolio-arkoprobho-mitra.vercel.app | GitHub: github.com/Arkoprobho-Mitra

PROFESSIONAL SUMMARY

AI Engineer and sub-team lead with 5+ years at Infosys, delivering production GenAI and ML systems for banking, insurance, and procurement. Led teams of 2-4 engineers across enterprise RAG, claims intelligence, and multi-agent automation. Strengths include secure retrieval, evaluation, observability, human approval controls, and cloud deployment.

TECHNICAL SKILLS

GenAI and evaluation: LangGraph, LangChain, RAG, Azure OpenAI, Azure AI Search, RAGAS, Langfuse

ML and data: Python, scikit-learn, XGBoost, SHAP, LayoutLMv2, MLflow, Evidently AI, PL/SQL, Elasticsearch

Engineering and cloud: FastAPI, asyncio, PostgreSQL, Docker, GitHub Actions, Azure Document Intelligence, Key Vault, Blob Storage, AWS EC2, SAP Ariba, ServiceNow, SharePoint

PROFESSIONAL EXPERIENCE

AI Engineer / Sub-team Lead | Infosys Topaz, Applied AI Practice | Kolkata, India | Nov 2023 - Present

Multi-Agent Procurement Copilot | Aug 2025 - Present

- Led a team of 3 AI engineers delivering a LangGraph-orchestrated procurement copilot for a global manufacturer; reported outcomes include a 35% reduction in procurement cycle time and RFQ turnaround reduced from 3 days to 4 hours.
- Designed a five-agent graph spanning planning, supplier, contract, compliance, and synthesis workflows; integrated SAP Ariba, ServiceNow, SharePoint, and contract RAG on Azure AI Search with access controls and blue-green index deployments.
- Established secretless access, human approval gates, and append-only audit logs to support governance and GDPR/DPA audit requirements; evaluated tool selection (over 90%), workflow completion (over 95%), RAGAS faithfulness, and SME scoring.

Insurance Claims Intelligence Platform | Nov 2024 - Sep 2025

- Led 4 AI engineers in delivering an end-to-end claims intelligence platform for a global insurer, with a reported 40% reduction in processing time across OCR, extraction, fraud detection, and human-review routing.
- Deployed Azure Document Intelligence and a LayoutLMv2 classifier with 94% accuracy on 15,000 labelled pages; built an explainable XGBoost fraud model using 800,000 claims with 3% fraud prevalence.
- Integrated Evidently AI drift monitoring with MLflow retraining workflows, reducing the retraining cycle from a five-day manual SLA to overnight execution after PSI threshold breaches.

Enterprise Knowledge Assistant - Global Banking | Nov 2023 - Dec 2024

- Led a two-engineer sub-team delivering a two-phase GenAI assistant over a 2.5 million-document corpus for a North American investment bank; the program reported a 60% reduction in compliance query time at 30,000+ monthly active users.
- Evolved the platform from custom Python orchestration to hybrid vector plus BM25 retrieval using Reciprocal Rank Fusion, Azure OpenAI, and LangChain, improving retrieval precision by 14% on regulation-code queries.
- Enforced role-based access with pre-retrieval Azure AD metadata filters; built a 200-query evaluation set with a compliance SME and embedded evaluation gates and RAGAS scoring in CI/CD.

EARLIER EXPERIENCE

Systems Engineer - Data and Logistics | Infosys | India | Aug 2021 - Oct 2023

- Optimized PL/SQL pipelines and Elasticsearch indices supporting daily logistics operations and real-time shipment search.
- Built QlikSense dashboards, maintained ServiceNow incident and change workflows, and supported data migrations and legacy-system integrations.
- Worked with cross-functional client teams on system integrations and operational data changes across logistics workflows.

RESEARCH PUBLICATION

How to handle bi/tri-lingual Indic texts in a single image? A new dataset of natural scene and born-digital images
Multimedia Tools and Applications, Springer, Vol. 81, pp. 15367-15394 (2022) | DOI: 10.1007/s11042-022-12596-7
Co-authored research that constructed and published a multilingual dataset of natural-scene and born-digital images containing Indic script combinations for OCR and scene-text detection in low-resource scripts.

SELECTED PROJECTS

Agentic Job Application Pipeline | 2026: LangGraph multi-agent workflow for job-listing ingestion, LLM scoring, tailored cover-letter drafting, and application-state tracking.

LangGraph Fact-Checker with Human Review | 2026: Multi-node graph with web search, claim verification, a human approval step, and Langfuse trace-level observability.

Full-stack E-Commerce Platform with RAG Assistant | 2025: Microservices-based commerce platform with semantic search, review sentiment classification, a RAG support assistant, and distributed authentication.

EDUCATION AND CERTIFICATION

Bachelor of Engineering (B.E.) | Jadavpur University, Kolkata, India | 2017 - 2021

AWS Certified Cloud Practitioner | Amazon Web Services